Cauchy Integral Formula

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of cauchy integral formula, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-faculty node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

CA-01 / Complex Analysis / Report v1

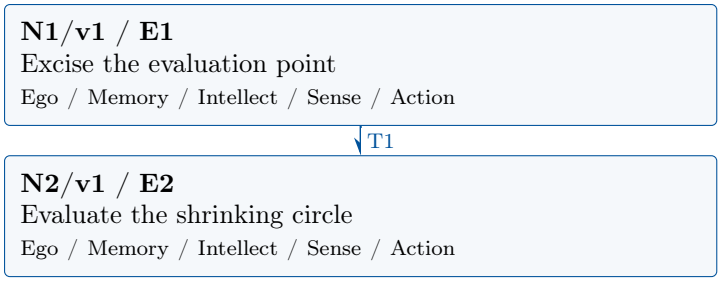
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let f be holomorphic on an open neighborhood of the closed disk $\overline{D(z_0, R)}$. For w inside the disk, prove $f(w) = \frac{1}{2\pi i} \oint_{|z-z_0|=R} f(z)/(z-w)\, dz$, with positive orientation.*

A. Foundations and permitted imports

Holomorphic means complex differentiable on an open neighborhood of the closed disk. Import Cauchy–Goursat in a finitely connected region with piecewise smooth boundary: the positively oriented boundary integral of a holomorphic function is zero. Also use continuity and the parametrization of a circle. Inner boundary components carry clockwise induced orientation.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five faculties. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N2. The outer integral equals the counterclockwise inner integral.

Receiving activity. N2: Evaluate the shrinking circle

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Equality of integrals; continuity at w; prior exports E1/v1

III. AGENT ROLES AND FIVE-FACULTY ACTIVITIES

A. N1/v1: Excise the evaluation point

Object and role. Move the integral to a small circle without crossing a singularity.

Inputs. w inside disk; f holomorphic near closure; foundational imports stated in §1

E — Ego

Own E1/v1: The outer integral equals the counterclockwise inner integral.

M — Memory

Retain w inside disk; f holomorphic near closure; foundational imports stated in §1; keep their assumptions and versions attached.

I — Intellect

Move the integral to a small circle without crossing a singularity.

S — Sense

The singular point must be removed; the sign of the inner boundary matters.

A — Action

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Choose $\rho > 0$ so the closed disk about w of radius ρ lies inside $D(z_0, R)$.

N1.2 On the region between the outer circle and this small disk, $h(z) = f(z)/(z-w)$ is holomorphic.

N1.3 Cauchy–Goursat with its induced boundary orientations gives $\oint$ outer h $dz - \oint |z-w| = \rho$ h $dz = 0$, where both displayed circles are individually counterclockwise.

Export E1/v1. The outer integral equals the counterclockwise inner integral.

Acceptance gate. The singular point must be removed; the sign of the inner boundary matters. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: The singular point must be removed; the sign of the inner boundary matters. Recheck dependent nodes N2.

B. N2/v1: Evaluate the shrinking circle

Object and role. Subtract the constant value and estimate the remainder.

Inputs. Equality of integrals; continuity at w; prior exports E1/v1

E — Ego

Own E2/v1: $f(w) = (2\pi i)^{-1}$ times the given boundary integral.

M — Memory

Retain Equality of integrals; continuity at w; prior exports E1/v1; keep their assumptions and versions attached.

I — Intellect

Subtract the constant value and estimate the remainder.

S — Sense

An integral bound uses $circumference 2\pi\rho$ and denominator ρ; they cancel.

A — Action

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Split $f(z) = f(w) + (f(z) - f(w))$.

N2.2 The constant part integrates to $2\pi i f(w)$, since $z = w + \rho e^{it}$ gives $dz/(z-w) = i$ dt.

N2.3 The remainder satisfies $|\oint (f(z) - f(w))/(z-w)\, dz| \le 2\pi \sup_{|} z - w| = \rho |f(z) - f(w)| \to 0$ by continuity.

N2.4 The outer integral is independent of ρ, so taking $\rho \to 0$ gives $\oint$ outer $f(z)/(z-w)dz = 2\pi i f(w)$, the desired formula.

Export E2/v1. $f(w) = (2\pi i)^{-1}$ times the given boundary integral.

Acceptance gate. An integral bound uses $circumference 2\pi\rho$ and denominator ρ; they cancel. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: An integral bound uses $circumference 2\pi\rho$ and denominator ρ; they cancel. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Excise the evaluation point

1. Choose $\rho > 0$ so the closed disk about w of radius ρ lies inside $D(z_0, R)$.
2. On the region between the outer circle and this small disk, $h(z) = f(z)/(z - w)$ is holomorphic.
3. Cauchy–Goursat with its induced boundary orientations gives $\oint$ outer h $dz - \oint |z-w| = \rho$ h $dz = 0$, where both displayed circles are individually counterclockwise.

B. §4.N2 — Evaluate the shrinking circle

1. Split $f(z) = f(w) + (f(z) - f(w))$.
2. The constant part integrates to $2\pi i f(w)$, since $z = w + \rho e^{it}$ gives $dz/(z - w) = i$ dt.
3. The remainder satisfies $|\oint (f(z)-f(w))/(z-w)\, dz| \leq 2\pi \sup_{|} z-w| = \rho|f(z)-f(w)| \to 0$ by continuity.
4. The outer integral is independent of ρ, so taking $\rho \to 0$ gives $\oint$ outer $f(z)/(z-w)dz = 2\pi i f(w)$, the desired formula.

□

V. WORKED ORCHESTRATION

For $f(z) = z^2 + 1$, $z_0 = 0, R = 2, w = 1$, the formula predicts $\oint (z^2+1)/(z-1)dz = 4\pi i$. Algebraically $z^2 + 1 = (z - 1)(z + 1) + 2$; the polynomial term integrates to 0 and $2/(z - 1)$ integrates to $4\pi i$.

A. Boundary and hypothesis audit

For w on the boundary the integrand is singular on the contour, and the excision argument is unavailable. Sense rejects the interior formula there; boundary limiting theorems require a different contract. Reversing the contour gives the negative of the stated integral.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Ego → Memory:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Intellect → Action:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Sense → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Memory + Intellect:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Action:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Ego:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

For w on the boundary the integrand is singular on the contour, and the excision argument is unavailable. Sense rejects the interior formula there; boundary limiting theorems require a different contract. Reversing the contour gives the negative of the stated integral.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- Harvard complex analysis: Cauchy formula, argument principle, and Riemann mapping lectures.